

Surya Krishna Bharadwaj Kunapuli

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Professional Summary

Final-year AI & CS student at Amrita Vishwa Vidyapeetham, graduating 2026. I build full-stack AI products and have published two papers at IEEE ICCNT-24 (IIT Mandi), one with 4 citations, plus a patent filed for a multimodal human-computer interaction system. Most of my work sits at the intersection of agent infrastructure and data systems — five products shipped and live, including a 28-table data governance platform and a multi-agent orchestration tool built in two days. My internship work has been more physical than people expect from an AI background: computer vision that watches for limbs near factory machinery, IoT sensors on warehouse equipment.

Patents

Hands-Free Multimodal Human-Computer Interaction System and Method Based on Vision and Speech Processing

Application No. 202641053085 · Filed Apr 2026 · Office of the Controller General of Patents, Designs & Trade Marks, India

Technical Skills

Languages: Python, TypeScript, JavaScript, Java

ML / AI: TensorFlow, Keras, Scikit-Learn, CNN, LSTM, BiLSTM, LightGBM, CTGAN, TVAE, Genetic Algorithms

Agent & LLM: Anthropic Claude API, Claude SDK, LangChain, Mem0, Groq, OpenAI, ONNX Runtime

Web & Mobile: Next.js 15, React 18/19, FastAPI, MERN Stack, Expo (React Native), WebSockets

Databases: PostgreSQL, Neon, Supabase, Drizzle ORM, SQLAlchemy, MongoDB

Infrastructure: Docker, Nginx, GitHub Actions CI/CD, Vercel, Railway, AWS, Azure, Cloudflare R2

Work Experience

AI Developer Intern — Paramount Computer Systems

Jan – May 2026 · Bengaluru

- Worked on an internal tool that automates support ticket triage across L1, L2, and L3 — it classifies tickets on arrival and routes them to the right tier, which cut down on the manual back-and-forth that used to happen at each handoff.
- Spent a good chunk of time on the classification logic itself, since getting ownership right at intake mattered more than I expected for actually reducing review time downstream.

AI Engineer Intern — almiT

Jun – Dec 2025 · Bengaluru

- Built the real-time computer vision module for Project Sonicster — it watches for human limbs near active production lines and triggers safety interlocks before contact happens. Had to keep alert latency under a second for it to be useful on an actual factory floor.
- Also built the IoT layer for remotely monitoring heavy machinery in warehouses, pulling in temperature, humidity, and other sensor data to catch equipment problems early rather than after they'd already caused downtime.

Personal Projects

Datrix — AI Data Infrastructure & Governance Platform · 2026 · FastAPI, React, Polars, PostgreSQL, CTGAN, Docker, GitHub Actions

- You upload raw files and Datrix takes it from there — quality scoring across five dimensions, a guided cleaning wizard, pipeline building, synthetic data generation when you don't have enough real data (CTGAN/TVAE), active learning, model benchmarking, and compliance reports for GDPR/HIPAA/CCPA. No code needed at any step.
- Underneath that: 28 SQLAlchemy tables, full auth with Google/GitHub OAuth, role-based approval workflows for teams, a Kanban board for change requests, S3/R2 storage, and a CI/CD pipeline through GitHub Actions. Live on Vercel.

Cortex — Memory-First Multi-Agent Orchestration · 2026 · React 18, FastAPI, Mem0, React Flow, Groq, Neon Postgres

- Most agent systems put the agents at the center and memory off to the side. I flipped that — the shared memory pool is the hub, and agents are just workers reading from and writing to it. Built the whole thing in about two days with Claude Code, 75 features and 36 bugs later, with every decision logged in TRANSCRIPTS.md as I went.
- Conflict detection happens in two stages — a cosine similarity check first, then a Groq LLM call if things look contradictory — surfaced in a visual panel where you can keep, merge, or dismiss. Three separate memory scopes (personal, org-shared, agent-private), real-time WebSocket events per agent.

AgentRank — Trust & Reputation Layer for AI Agents · 2026 · Next.js 15, TypeScript, PostgreSQL, Drizzle ORM, Auth.js v5

- MCP and A2A handle how agents connect to each other, but neither tells you whether an agent is actually any good. AgentRank does — builders publish profiles, agents move through trust tiers (Bronze to Platinum) based on real reviews, and you can search and compare instead of digging through forums. GitHub OAuth keeps onboarding frictionless.

GlassBox-fin — Compliance Guardrails for Financial LLM Agents · 2026 · *FastAPI, Python, React, D3, PostgreSQL, SSE*

- Compliance middleware that wraps around any agent framework. Eight built-in rules covering GDPR, the EU AI Act, MiFID II, and Basel III, with hash-chain audit trails so nothing can be quietly altered after the fact. Flagged decisions go through human approval before anything happens, and there's a live trace monitor in the control plane.

Grain — User Feedback Intelligence for Product Teams · 2026 · *React 19, TypeScript, Supabase, TanStack Query v5*

- Built around one idea: a feature request is usually a symptom, not the actual problem. So Grain forces PMs to log the underlying issue, then clusters similar signals automatically. Stakeholders get share links with no login required, and everything's isolated per workspace through Supabase row-level security. Deployed on Vercel.

Academic Projects

Multi-Tier Immune-Based Adaptive Defence Model for DDoS Detection

Oct 2025

- Modeled the defense layers on the human immune system — rule-based filters catch the obvious stuff, entropy detection flags anomalies, then CNN-LSTM/BiLSTM classifiers and a LightGBM-autoencoder hybrid handle what's left. A genetic algorithm tunes the hyperparameters, and the memory layer adapts to new attack patterns without needing a full retrain each time.

Publications

Dynamic Resource Allocation for Edge Computing · IEEE ICCCNT-24, IIT Mandi, Jun. 2024 · 4 citations

Combined quantum-inspired algorithms with game theory for multi-agent coordination, testing it on edge environments with tight resource constraints. Beat static baselines on both latency and resource use.

Dynamic Portfolio Optimization Using Proximal Gradient Method · IEEE ICCCNT-24, IIT Mandi, Jun. 2024

Used proximal gradient descent to handle the non-smooth L1 constraints that standard methods struggle with in portfolio optimization. Ran faster on real historical stock data while still respecting those constraints.

DOI: [10.1109/ICCCNT61001.2024.10724361](https://doi.org/10.1109/ICCCNT61001.2024.10724361)

Education

B.Tech — Artificial Intelligence & Computer Science — [Amrita Vishwa Vidyapeetham, Bengaluru](#)

2022 – 2026

- Published twice at IEEE ICCCNT-24, IIT Mandi (one paper has 4 citations so far), and have a patent on file. Most of my coursework-adjacent research has centered on multi-agent systems and resource allocation under uncertainty — themes that show up again in the products I build outside of class.

XII — MPC (CBSE) — [Sri Chaitanya Techno School](#)

2022 · 89.4%

X — CBSE — [Sri Chaitanya Techno School](#)

2020 · 95.4%

Certifications

AI for All: Basics to GenAI Practice · NVIDIA Academy, 2024

Generative AI for Everyone · DeepLearning.AI, Coursera, 2024